WORKED SOLUTION

STEP II 2002, Question 7

Pure • Vector Geometry of Lines

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s2-q7

The key idea

Everything here is a dot product. A line through the origin with unit direction $\mathbf{u}$ makes an angle of $\pi/4$ with m_1 exactly when $|\mathbf{u}\cdot\mathbf{d}_1| = |\mathbf{d}_1|\cos\frac{\pi}{4}$, which turns the first part into simultaneous equations. For the second half, "perpendicular" is a dot product equal to zero, and two lines in space meet only if they are coplanar and not parallel; coplanarity is a scalar triple product equal to zero. The twist is that the coplanar case turns out to be a parallel one.

Solution

The lines m_3 and m_4

Let $\mathbf{d}_1 = \mathbf{i}+\mathbf{j}$ and $\mathbf{d}_2 = \mathbf{i}+\mathbf{k}$, both of length $\sqrt{2}$. Let $\mathbf{u} = a\mathbf{i}+b\mathbf{j}+c\mathbf{k}$ be a unit vector along a line through O. The angle between two lines is the acute angle between them, so the condition is

$$|\mathbf{u}\cdot\mathbf{d}_1| = \sqrt{2}\cos\tfrac{\pi}{4} = 1, \qquad |\mathbf{u}\cdot\mathbf{d}_2| = 1,$$

that is, $a+b = \pm 1$ and $a+c = \pm 1$. Replacing $\mathbf{u}$ by $-\mathbf{u}$ gives the same line and flips both signs, so there are only two cases.

Same signs: $a+b=1$, $a+c=1$. Then $b=c=1-a$, and $a^2+b^2+c^2=1$ gives

$$a^2+2(1-a)^2 = 1 \iff 3a^2-4a+1=0 \iff (3a-1)(a-1)=0.$$

So $a=1$, giving $\mathbf{u}=\mathbf{i}$, or $a=\frac{1}{3}$, giving $\mathbf{u}=\frac{1}{3}(\mathbf{i}+2\mathbf{j}+2\mathbf{k})$.

Opposite signs: $a+b=1$, $a+c=-1$. Then $b=1-a$, $c=-1-a$, and $a^2+(1-a)^2+(1+a)^2=1$ gives $3a^2+2=1$, which has no real solution.

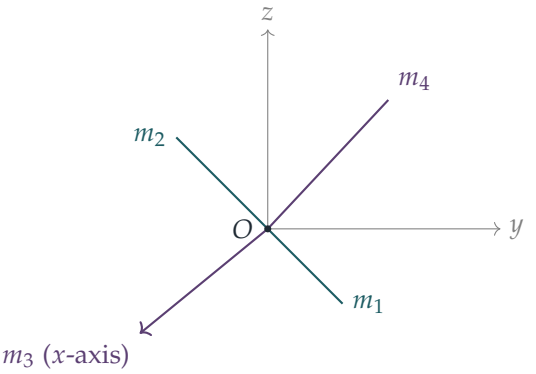
So the two lines have directions

$$m_3:\ \mathbf{i}, \qquad m_4:\ \mathbf{i}+2\mathbf{j}+2\mathbf{k}$$

(the labels could equally be swapped). With unit vectors $\mathbf{e}_3=\mathbf{i}$ and $\mathbf{e}_4=\frac{1}{3}(\mathbf{i}+2\mathbf{j}+2\mathbf{k})$,

$$\cos\theta = \mathbf{e}_3\cdot\mathbf{e}_4 = \tfrac{1}{3},$$

which is positive, so this is the acute angle: its cosine is $\frac{1}{3}$.



Setting up the points

Since A lies on m_1 at distance $\lambda\sqrt{2}$ from O, it is $\pm\lambda(1,1,0)$; non-negative coordinates force us to take $A = \lambda(1,1,0)$ with $\lambda \geq 0$ (we may take $\lambda \geq 0$ throughout, since $\lambda\sqrt{2}$ is a distance). Likewise $B = \lambda(1,0,1)$. The unit points on m_3, m_4 with non-negative coordinates are $(1,0,0)$ and $\frac{1}{3}(1,2,2)$.

Which of these is P and which is Q depends on the labelling of m_3 and m_4. It doesn't matter: the reflection $(x,y,z) \mapsto (x,z,y)$ swaps A and B and fixes both $(1,0,0)$ and $\frac{1}{3}(1,2,2)$, so it carries one labelling of the configuration onto the other (with the roles of the lines AQ and BP exchanged). Perpendicularity and intersection are preserved by a reflection, so it is enough to take

$$A = (\lambda,\lambda,0), \quad B = (\lambda,0,\lambda), \quad P = (1,0,0), \quad Q = (\tfrac{1}{3},\tfrac{2}{3},\tfrac{2}{3}).$$

Then

$$\overrightarrow{AQ} = (\tfrac{1}{3}-\lambda,\ \tfrac{2}{3}-\lambda,\ \tfrac{2}{3}), \qquad \overrightarrow{BP} = (1-\lambda,\ 0,\ -\lambda).$$

Part (i)

AQ is perpendicular to BP exactly when $\overrightarrow{AQ} \cdot \overrightarrow{BP} = 0$:

$$\begin{aligned}\overrightarrow{AQ} \cdot \overrightarrow{BP} &= (\tfrac{1}{3}-\lambda)(1-\lambda) + 0 - \tfrac{2}{3}\lambda \\ &= \lambda^2 - \tfrac{4}{3}\lambda + \tfrac{1}{3} - \tfrac{2}{3}\lambda = \lambda^2 - 2\lambda + \tfrac{1}{3}.\end{aligned}$$

This vanishes when $(\lambda-1)^2 = \frac{2}{3}$, that is,

$$\lambda = 1 \pm \sqrt{\tfrac{2}{3}} = 1 \pm \tfrac{\sqrt{6}}{3}.$$

Both values are positive (since $\sqrt{6} < 3$), so both are allowed by the non-negativity condition, and the quadratic has no other roots. Hence there are exactly two values of λ for which $AQ \perp BP$, as required.

Part (ii)

Ask yourself How do we test whether two lines in three dimensions meet?

Two distinct lines meet if and only if they lie in a common plane and are not parallel. The lines AQ and BP lie in a common plane exactly when the four points A, B, P, Q do, which happens exactly when the scalar triple product of $\overrightarrow{AB}$, $\overrightarrow{AP}$, $\overrightarrow{AQ}$ is zero.

Here $\overrightarrow{AB} = (0, -\lambda, \lambda)$ and $\overrightarrow{AP} = (1-\lambda, -\lambda, 0)$, so

$$\begin{vmatrix} 0 & -\lambda & \lambda \\ 1-\lambda & -\lambda & 0 \\ \frac{1}{3}-\lambda & \frac{2}{3}-\lambda & \frac{2}{3} \end{vmatrix} = \lambda \cdot \tfrac{2}{3}(1-\lambda) + \lambda\Big[(1-\lambda)(\tfrac{2}{3}-\lambda) + \lambda(\tfrac{1}{3}-\lambda)\Big]$$
$$= \lambda\Big[\tfrac{2}{3} - \tfrac{2}{3}\lambda + \tfrac{2}{3} - \tfrac{4}{3}\lambda\Big] = 2\lambda(\tfrac{2}{3} - \lambda),$$

expanding along the first row. For $\lambda \neq 0$, the lines can therefore only meet if $\lambda = \frac{2}{3}$.

But when $\lambda = \frac{2}{3}$,

$$\overrightarrow{AQ} = (-\tfrac{1}{3},\ 0,\ \tfrac{2}{3}), \qquad \overrightarrow{BP} = (\tfrac{1}{3},\ 0,\ -\tfrac{2}{3}) = -\overrightarrow{AQ},$$

so the lines are parallel. They are not the same line, because $\overrightarrow{BA} = (0, \frac{2}{3}, -\frac{2}{3})$ is not a multiple of $\overrightarrow{BP}$, so A is not on BP. Two distinct parallel lines never meet.

So for every non-zero λ the lines AQ and BP fail to intersect, as required. (When $\lambda = 0$, $A = B = O$ and the two lines do meet at O, which is why the question excludes it.)

Checks

- $\mathbf{i}$ makes angle $\pi/4$ with $\mathbf{i} + \mathbf{j}$ by inspection, which is a reassuring first root of the quadratic.
- At $\lambda = 1 + \sqrt{2/3} \approx 1.8165$ the dot product $\overrightarrow{AQ} \cdot \overrightarrow{BP}$ evaluates numerically to 0 (to 10^{-16}), and likewise at $\lambda \approx 0.1835$.
- Computing the shortest distance between the lines AQ and BP for $\lambda = 0.001, 0.002, \ldots, 5$ (and for the other labelling of P and Q) never gives zero; it only tends to zero as $\lambda \to 0$.

Common slips

- Forgetting the case $a + b = 1$, $a + c = -1$: it gives nothing, but it has to be ruled out.
- Stopping at "the triple product vanishes when $\lambda = \frac{2}{3}$, so they intersect". Coplanar lines can be parallel, and here they are.

Beyond STEP

The scalar triple product is a 3×3 determinant, and its vanishing says that three vectors are linearly dependent; the shortest distance between two skew lines, $|\overrightarrow{AB} \cdot (\mathbf{u} \times \mathbf{v})|/|\mathbf{u} \times \mathbf{v}|$, is built from the same pieces. In a first-year linear algebra course these ideas are unified through determinants, rank and the geometry of subspaces. Strang's *Introduction to Linear Algebra* develops this with plenty of three-dimensional pictures.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.